

Task 1 Report: Data Understanding and Related Work

GNN-Based EMG Gesture Recognition on NinaPro DB1

Group 04 | Track 1 — Graph Neural Network | East West University

1. Dataset Overview

This project uses NinaPro DB1, a public surface EMG dataset recorded from 10 forearm electrodes while subjects performed a series of hand and finger gestures. Each timepoint is labelled with the gesture being performed (stimulus/restimulus), the repetition number, and the subject ID.

Property	Value
Dataset	NinaPro DB1
Source	ninapro.hevs.ch (Kaggle: mansibmursalin/ninapro-db1-full-dataset)
Application area	EMG gesture recognition
EMG channels	10
Missing values	0
Duplicate rows	0
Track	Track 1 — GNN on table-style data

2. Feature Extraction

Raw EMG timepoints carry little individual meaning, since a gesture is expressed as a pattern over time rather than a single instant. The continuous signal was segmented into overlapping windows (200 samples, 100-sample step), and six standard time-domain descriptors were computed per channel per window: RMS, MAV, WL, VAR, ZC, and SSC. This produced the tabular feature set used for the EDA below.

3. Exploratory Data Analysis — Key Findings

- Class balance: the "rest" class dominates, outnumbering any single gesture class by roughly 30–80×, motivating macro-F1 and per-class recall over raw accuracy.
- Data quality: no missing values or duplicates; ZC is near-constant (uninformative) on channels 2, 4, and 7.
- Redundancy: MAV is highly correlated (>0.95) with RMS on nearly every channel, and WL overlaps with both on several channels — candidates for removal before modeling.
- Distributions: RMS/MAV/VAR features are strongly right-skewed, reflecting mostly low-activity windows with occasional high-amplitude contraction bursts.
- Separability: PCA/t-SNE show partial, non-linear class separation, suggesting a linear classifier alone is unlikely to be sufficient — supporting the case for a GNN.
- Subject structure: window counts vary by subject and class, confirming a subject-wise (grouped) train/test split is required to avoid data leakage.

4. Related Work

Ten studies (2022–2026) on GNN/GCN-based EMG and physiological-signal gesture recognition were reviewed:

Authors (Year)	Application Area	GNN Type	Key Limitation / Gap
Xu, Chen, Ruan, & Zhang (2023)	Cross-user EMG gesture recognition	Spatio-temporal GCN	Fixed channel-graph topology

Lee et al. (2023)	Static/dynamic gesture recognition	GNN on flexible-array electrodes	Custom hardware, not comparable to public datasets
Massa, Riboni, & Nazarpour (2023)	Fine-grained gesture-intention recognition	GNN + GNNExplainer	High error variance; real-time cost
Lee, Jiang, Yang, Yang, & Zhao (2025)	Generalizable gesture classification	Spatiotemporal GCN	Prioritizes cross-dataset generalization
Massa, Riboni, & Nazarpour (2022)	Movement-intention recognition	Baseline GNN	Simple, non-adaptive graph topology
Vipulanandan, Premaratne, & Murthi (2024)	Real-time gesture recognition	GNN, correlation-based graph	Only 5 gesture classes tested
Li et al. (2023)	Multimodal stroke rehabilitation BCI	GIN, heterogeneous graph	Added complexity vs. EMG-only setups
Quadrelli et al. (2025)	Prosthetic control (review)	Review of GNN/spatial ML	Review scope; no single-model benchmark
Klepl, He, Wu, Blackburn, & Sarriani (2024)	EEG classification survey	Survey of GNN variants	EEG-focused, not low-channel sEMG
Hou, Jia, Lun, Hao, Shi, Li, Zeng, & Lv (2022)	EEG motor-imagery decoding	GCNs-Net	Designed for EEG, not sparse EMG

5. Research Gap

- Most GNN-based EMG work targets high-density (HD) sEMG with dozens of electrodes; low-channel (10-electrode) datasets like NinaPro DB1 remain largely unexamined with GNNs.
- Graph topology in prior work is typically fixed by physical electrode adjacency rather than learned from inter-channel signal correlation — the more relevant cue with only 10 channels.
- Custom-hardware studies report strong accuracy but on non-public datasets, limiting reproducibility.
- Severe rest-vs-gesture class imbalance is rarely addressed explicitly within the graph design or training strategy itself.

This work addresses these gaps by constructing a correlation-driven channel graph from a standard low-density NinaPro configuration, evaluated under a subject-independent, leakage-free split with explicit class-imbalance handling.

6. Summary

The EDA confirms that NinaPro DB1 requires windowed feature extraction, contains redundant amplitude-related features, and exhibits severe class imbalance dominated by the rest class. Combined with the identified research gap, these findings directly inform the correlation-based channel graph and evaluation protocol for the GNN model to be developed in Task 2.